MINI-PROJECT BLUEPRINT & DESIGN DOCUMENT

# SignVaani

## Real-Time Indian Sign Language (ISL) to Multilingual Speech Translator

Keypoint-based recognition | Custom ISL dataset | Offline Indic speech output

Domain: Computer Vision, Deep Learning, NLP, Assistive Technology

Project Type: B.Tech / B.E. Mini-Project (8 weeks, team of 2-4)

Submitted by: _____ Roll No: _____

Guide: _____ Department: _____

Institution: _____

*Version 1.0 | October 2026*

# Contents


- 1. Abstract
- 2. Introduction & Problem Statement
- 3. Objectives
- 4. Scope (In / Out)
- 5. Literature Survey & Existing Systems
- 6. Proposed System & Architecture
- 7. Technology Stack (with justification)
- 8. Datasets & Data Collection Plan
- 9. Methodology: Module-by-Module Implementation
- 10. What Makes This Project Unique
- 11. Evaluation Plan & Metrics
- 12. Hardware & Software Requirements
- 13. Project Folder Structure
- 14. 8-Week Timeline & Team Roles
- 15. Risks & Mitigation
- 16. Ethics, Privacy & Accessibility
- 17. Expected Outcomes & Deliverables
- 18. Future Scope
- 19. Conclusion
- 20. References
- Appendix A: Suggested vocabulary (Hospital domain, 40 signs)
- Appendix B: Viva / review questions

# 1. Abstract

India has an estimated 6 crore+ people with hearing disabilities, yet only a few hundred certified Indian Sign Language (ISL) interpreters. Most sign-language recognition research and tools target American Sign Language (ASL); ISL has distinct vocabulary, uses two hands for many signs, and has its own grammar, so ASL tools do not transfer. **SignVaani** is a real-time system that recognises isolated ISL signs from a normal webcam, assembles them into a grammatical sentence, and speaks the sentence aloud in English, Hindi or a regional language. Instead of processing raw video, the system extracts 3D body, hand and face landmarks using MediaPipe Holistic and classifies landmark sequences with a lightweight recurrent / transformer model, allowing real-time inference on an ordinary laptop CPU without internet. The project also contributes a small, documented, signer-diverse ISL dataset for a chosen domain (e.g. hospital communication) and reports signer-independent accuracy, which most student projects omit.

**Keywords:** Indian Sign Language, MediaPipe, keypoint sequence classification, GRU, Transformer, text-to-speech, assistive technology, low-resource languages.

# 2. Introduction & Problem Statement

Deaf and hard-of-hearing people in India face daily communication barriers in hospitals, banks, railway counters and government offices, where staff do not know ISL and interpreters are rarely available. Writing notes is slow and assumes literacy in a spoken language, which many ISL users acquired late.

**Problem statement:** *Design and implement a low-cost, real-time, offline-capable system that converts a defined vocabulary of Indian Sign Language signs, performed in front of a standard webcam, into grammatical spoken sentences in English and Indian languages, and evaluate it on signers not seen during training.*

### *Why is this hard?*

- **Low resource:** public ISL datasets are small compared to ASL (WLASL, MS-ASL).
- **Two-handed and non-manual features:** facial expression, mouth shape and head movement change meaning.
- **Grammar mismatch:** ISL word order (often topic-comment, verb-final, e.g. ME HOSPITAL GO) differs from English and Hindi.
- **Regional variation:** the same concept can be signed differently in Mumbai, Delhi or Hyderabad.
- **Real-time segmentation:** the system must decide when a sign starts and ends in a continuous stream.

# 3. Objectives

- Build a pipeline that recognises **30 to 50 isolated ISL signs** from one domain in real time (target: at least 15 FPS, under 300 ms latency per sign on a laptop CPU).
- Create and publish a **custom ISL keypoint dataset** (target: 20 to 30 clips per sign from at least 6 signers).
- Achieve **at least 85% top-1 accuracy on unseen signers** (signer-independent split) for the chosen vocabulary.
- Convert a sequence of recognised glosses into a grammatical English sentence and translate it into at least one Indian language.
- Speak the sentence aloud using an **offline** TTS engine.

- Provide a clean UI with live skeleton overlay, captions, confidence scores and a correction option.
- Validate usability with at least one real ISL user or interpreter.

## 4. Scope

| In scope (will be delivered) | Out of scope (stated honestly) |
|---|---|
| Isolated sign recognition for 30-50 signs from one domain | Open-vocabulary continuous ISL translation |
| Sign spotting in a live stream using an idle class + motion gating | Fingerspelling of arbitrary names (optional stretch goal: A-Z static alphabet) |
| Gloss-to-sentence conversion (rules + optional LLM) | Full ISL grammar parsing (classifiers, spatial reference) |
| Speech in English + Hindi (+1 regional language) | Speech-to-sign (reverse direction) - listed as future scope |
| Laptop (offline) app; optional browser demo | Production-grade mobile app |
| Signer-independent evaluation with 6+ signers | Large-scale clinical validation |

# 5. Literature Survey & Existing Systems

Sign language recognition (SLR) approaches have moved from sensor gloves, to hand-crafted image features, to CNNs on raw video, to pose / keypoint-based models. Keypoint-based models are now favoured for low-resource, real-time settings because they are small, robust to background and lighting, and protect privacy (no raw video needs to be stored).

| Work / System | Approach | Limitation addressed by this project |
|---|---|---|
| Sensor-glove systems | Flex sensors + accelerometer on a glove | Expensive, intrusive, ignore face and body |
| CNN on RGB frames (e.g. I3D, 3D-CNN) | Learn from raw video | Needs GPU and lots of data; sensitive to background |
| INCLUDE (AI4Bharat, 2020) | Word-level ISL dataset (~4,000 videos, 263 signs) + pose-based baselines | Lab-recorded; no speech output; no domain sentence building |
| OpenHands (AI4Bharat) | Pose-based SLR library, pretrained models for several sign languages incl. ISL | Research toolkit, not an end-user app |
| ISL-CSLTR | Sentence-level ISL corpus | Small; limited signers |
| iSign (2024) | Large ISL-English benchmark from online videos | Continuous translation still low accuracy; heavy models |
| Google MediaPipe + LSTM tutorials | Keypoints + LSTM, mostly ASL, few signs | Usually signer-dependent evaluation, inflated accuracy |
| Commercial ASL apps | ASL only | Do not support ISL vocabulary or Indian languages |

**Research gap:** there is no lightweight, offline, domain-focused ISL-to-speech tool that (a) uses face and body as well as hands, (b) outputs grammatical sentences in Indian languages, and (c) honestly reports signer-independent accuracy. SignVaani targets exactly this gap.

*Note: verify the latest versions, sizes and licences of INCLUDE, ISL-CSLTR and iSign before use and cite them correctly.*

# 6. Proposed System & Architecture

The system is a modular pipeline. Every stage can be tested and replaced independently, which also makes it easy to divide work among team members.

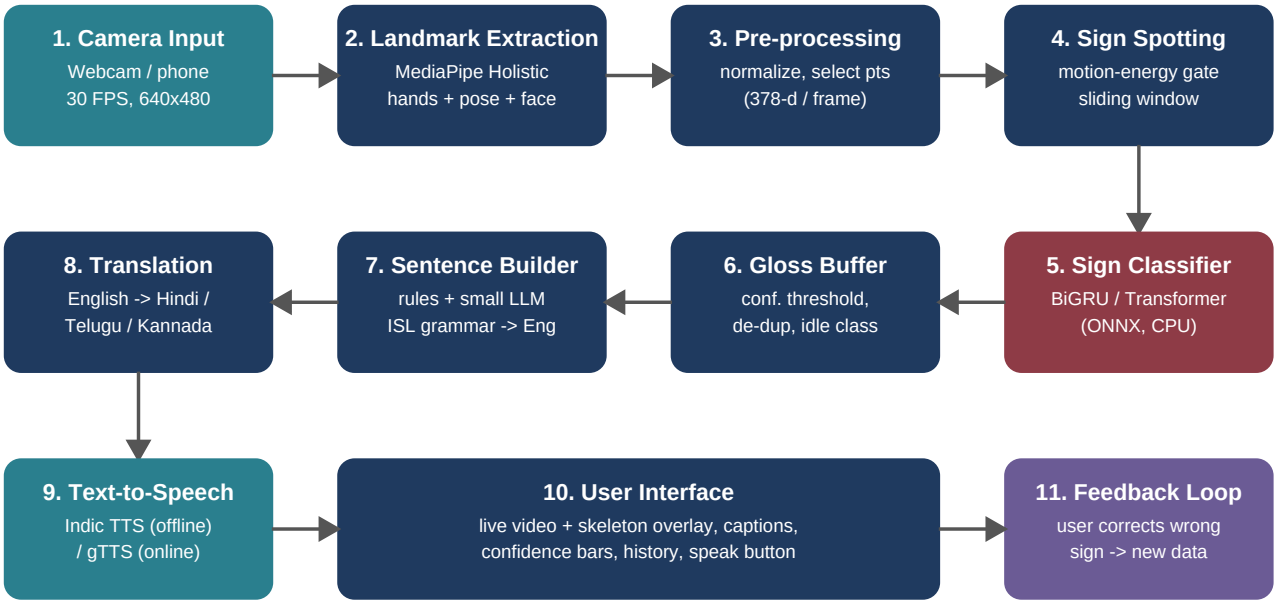


*Figure 1: End-to-end pipeline (all stages run locally on a laptop CPU)*

## Stage summary

| # | Stage | Input | Output | Key idea |
|---|---|---|---|---|
| 1 | Capture | Webcam | BGR frames | OpenCV VideoCapture, fixed 30 FPS |
| 2 | Landmarks | Frame | 33 pose + 2x21 hand + selected face pts | MediaPipe Holistic / Tasks API |
| 3 | Pre-processing | Raw landmarks | 378-d vector (63 pts x 3 + velocity) | Centre on shoulders, scale by shoulder width, fill missing hands with 0 + mask |
| 4 | Sign spotting | Vector stream | Candidate window (30-45 frames) | Hand-motion energy + idle class decide start/end |
| 5 | Classifier | T x 378 window | Sign + probability | BiGRU baseline; Transformer encoder as improvement |
| 6 | Gloss buffer | Predictions | Gloss list | Accept if p > 0.8 for k frames; suppress repeats |
| 7 | Sentence builder | ME PAIN STOMACH | I have stomach pain. | Templates first, small LLM optional |
| 8 | Translation | English | Hindi / Telugu / Kannada | IndicTrans2 (offline) or rule tables |
| 9 | TTS | Text | Audio | AI4Bharat Indic-TTS / pyttsx3 / gTTS |
| 10 | UI | All | Screen + audio | Streamlit or FastAPI + JS frontend |

# 7. Technology Stack (with justification)

| Layer | Recommended | Alternatives | Why this choice |
|---|---|---|---|
| Language | Python 3.10/3.11 | - | All CV/ML libraries are Python-first |
| Video capture | OpenCV 4.x | imageio, browser getUserMedia | Simple, fast, cross-platform |
| Landmarks | MediaPipe Holistic (or MediaPipe Tasks: Pose + Hand + Face Landmarker) | OpenPose, MMPose, RTMPose | Runs real-time on CPU; gives hands, body and face together |
| Deep learning | PyTorch 2.x | TensorFlow / Keras | Clean code for custom sequence models; easy ONNX export |
| Models | BiGRU (baseline), Transformer encoder, 1D-CNN/TCN | SPOTER, OpenHands SL-GCN | Small (under 2M params), fast, good on keypoints |
| Experiment tracking | Weights & Biases or TensorBoard | MLflow | Compare runs, log confusion matrices |
| Data handling | NumPy, pandas, scikit-learn | - | Splits, metrics, label encoding |
| Deployment format | ONNX Runtime (CPU) | TFLite, TorchScript | 2-5x faster CPU inference; mobile path |
| Sentence generation | Rule-based templates + optional small LLM (via Ollama, e.g. a 1-3B model) | Cloud LLM API | Offline, predictable; LLM only to polish grammar |
| Translation | AI4Bharat IndicTrans2 | Google Translate API | Open source, offline, 22 Indian languages |
| Text-to-speech | AI4Bharat Indic-TTS / Coqui TTS | gTTS (online), pyttsx3 (English, offline) | Natural Indian-language voices |
| UI | Streamlit + streamlit-webrtc | FastAPI + WebSocket + HTML/JS; Gradio | Fast to build; demo-ready |
| Version control | Git + GitHub | - | Team work, releases |
| Dataset hosting | Kaggle / Hugging Face Datasets | Zenodo | Free, citable, DOI possible |

## Framework decision in one line

**MediaPipe (perception) + PyTorch (learning) + ONNX Runtime (inference) + IndicTrans2/Indic-TTS (language and speech) + Streamlit (interface).** All components are free, open source and work offline.

# 8. Datasets & Data Collection Plan

| Source | Type | How it is used |
|---|---|---|
| INCLUDE / INCLUDE-50 (AI4Bharat) | Word-level ISL videos | Pre-training, plus extra samples for signs that overlap your vocabulary |
| ISL-CSLTR | Sentence-level ISL | Testing gloss-to-sentence on real ISL sentences |
| iSign | ISL-English pairs | Reference for ISL grammar patterns; stretch goal |
| Own recordings (core contribution) | 30-50 domain signs, 6+ signers | Main training and signer-independent testing |

## Recording protocol

- Learn signs from a reliable source (ISLRTC dictionary videos / a local deaf school or interpreter) and get them checked.
- Signers: at least 6 people, varied gender, skin tone, hand size, left/right-handed; ideally at least 1-2 native ISL users.
- Per sign per signer: 5 clips, 2-3 seconds each, 30 FPS; vary background, lighting, distance (1-2 m) and clothing.
- Record an extra **IDLE** class (hands resting, scratching face, adjusting hair) - essential to prevent false triggers.
- Store only keypoints (.npy) in the public release; keep raw video private unless consent allows sharing.
- Use a simple recorder script that shows the sign name, a 3-2-1 countdown, and auto-saves files as `sign/signerID_take.npy`.
- Collect written consent; document everything in a **datasheet** (who, how, licence such as CC BY 4.0).

## Data augmentation (on keypoints)

- Small random rotation (+/- 15 deg), scaling (0.9-1.1), translation
- Temporal: speed change (0.8x-1.2x) by resampling, random frame drop, random crop of start/end
- Gaussian noise on coordinates (sigma about 0.01)
- Mirroring left/right (only if the sign is symmetric - check with an ISL user)
- Hand drop-out: randomly zero one hand for a few frames to simulate tracking failures

# 9. Methodology: Module-by-Module Implementation

## 9.1 Keypoint extraction

MediaPipe Holistic gives 33 pose, 21 left-hand, 21 right-hand and 468 face landmarks per frame. Using all 1,662 values is wasteful, so we keep upper-body pose (shoulders, elbows, wrists, nose, eyes), both hands, and about 20 face points (lips, eyebrows, eye corners) that carry non-manual meaning.

```
import cv2, numpy as np, mediapipe as mp
mp_h = mp.solutions.holistic
POSE_IDX = [0, 2, 5, 11, 12, 13, 14, 15, 16]                # nose, eyes, shoulders, elbows, wrists
FACE_IDX = [61, 291, 0, 17, 13, 14, 70, 105, 300, 334, 33, 263]  # lips, brows, eyes

def extract(results):
    def arr(lms, idx=None, n=21):
        if lms is None: return np.zeros((len(idx) if idx else n, 3))
        pts = np.array([[l.x, l.y, l.z] for l in lms.landmark])
        return pts[idx] if idx else pts
    pose = arr(results.pose_landmarks, POSE_IDX)
    face = arr(results.face_landmarks, FACE_IDX)
    lh, rh = arr(results.left_hand_landmarks), arr(results.right_hand_landmarks)
    return np.concatenate([pose, face, lh, rh])              # (63, 3)

with mp_h.Holistic(model_complexity=1) as holistic:
    cap = cv2.VideoCapture(0)
    while cap.isOpened():
        ok, frame = cap.read()
        res = holistic.process(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
        kp = extract(res)        # append to a buffer / save as .npy
```

## 9.2 Normalisation (signer-independence)

Raw coordinates depend on where the person stands and how big they appear. We translate all points so the mid-point of the shoulders is the origin, divide by shoulder width, and express each hand also relative to its own wrist (giving hand-shape features independent of hand position). Velocity features (frame differences) are appended, which helps distinguish signs with the same hand shape but different motion.

```
def normalize(seq):                        # seq: (T, 63, 3)
    ls, rs = seq[:, 3], seq[:, 4]          # left / right shoulder (indices in POSE_IDX order)
    centre = (ls + rs) / 2
    width = np.linalg.norm(ls - rs, axis=-1).mean() + 1e-6
    seq = (seq - centre[:, None]) / width
    vel = np.diff(seq, axis=0, prepend=seq[:1])
    return np.concatenate([seq, vel], -1).reshape(len(seq), -1)   # (T, 378)

def resample(seq, T=32):                   # fixed length via linear interpolation
    idx = np.linspace(0, len(seq) - 1, T)
    return np.stack([seq[int(np.floor(i))] for i in idx])
```

## 9.3 Sign classifier

*Figure 2: Baseline sign classifier (N signs + 1 'idle / no-sign' class)*

```
import torch, torch.nn as nn
class SignGRU(nn.Module):
    def __init__(self, in_dim=378, n_classes=41, h=128):
        super().__init__()
        self.proj = nn.Sequential(nn.Linear(in_dim, 256), nn.LayerNorm(256), nn.GELU(), nn.Dropout(0.2))
```

```
        self.gru  = nn.GRU(256, h, num_layers=2, batch_first=True, bidirectional=True, dropout=0.3)
        self.att  = nn.Linear(2 * h, 1)
        self.head = nn.Linear(2 * h, n_classes)
    def forward(self, x):                      # x: (B, T, in_dim)
        z, _ = self.gru(self.proj(x))
        w = torch.softmax(self.att(z), dim=1)  # attention over time
        return self.head((w * z).sum(1))
```

| Hyper-parameter | Value |
|---|---|
| Sequence length T | 32 frames (about 1 s); try 24 / 48 |
| Optimiser | AdamW, lr 1e-3, weight decay 1e-2, cosine schedule |
| Loss | Cross-entropy with label smoothing 0.1 |
| Batch / Epochs | 64 / 80-150 with early stopping on validation signers |
| Improvement models | Transformer encoder (4 layers, 4 heads, d=192); 1D-CNN/TCN; ensemble |

## 9.4 Real-time sign spotting

- Keep a rolling buffer of the last 45 frames.
- Compute hand motion energy (mean wrist speed). When it rises above a threshold, start capturing; when it stays low for about 8 frames, close the window.
- Classify the captured window. Accept the sign only if probability > 0.80 and the top class is not IDLE.
- Add a short cool-down (0.5 s) and ignore an identical consecutive sign unless the hands returned to rest in between.
- Show a progress ring in the UI so users know when the system is 'listening'.

## 9.5 Gloss-to-sentence generation

ISL glosses lack articles, tense markers and often use a different order. A two-tier approach is used:

- **Tier 1 - rules / templates:** a JSON grammar per domain, e.g. pattern [PRONOUN, SYMPTOM, BODYPART] -> "{pronoun_subject} have {bodypart} {symptom}." Handles pronoun mapping (ME->I), question markers (WHERE, WHAT + raised eyebrows -> question mark), negation (NOT / head shake).
- **Tier 2 - optional LLM polish:** a small local model receives the glosses plus a few-shot prompt and returns one short sentence. Its output is constrained to use only the recognised concepts, preventing invented content.
- Show both the gloss sequence and the generated sentence so the user can confirm before speaking.

```
RULES = {("ME", "PAIN", "*"): "I have {2} pain.",
         ("ME", "NEED", "*"): "I need {2}.",
         ("WHERE", "*"):      "Where is the {1}?"}
# "ME PAIN STOMACH" -> "I have stomach pain."   "WHERE TOILET" -> "Where is the toilet?"
```

## 9.6 Translation and speech

The English sentence is translated with IndicTrans2 (offline) and spoken with Indic-TTS. For a fixed domain, a pre-translated phrase table (verified by a native speaker) is faster and more reliable; the neural translator is used only as a fallback. Audio is cached so repeated phrases play instantly.

## 9.7 User interface

- Live video with skeleton overlay and a 'listening' indicator
- Top-3 predictions with confidence bars

- Gloss strip -> sentence -> Speak button (or auto-speak mode)
- Language selector (English / Hindi / Telugu / Kannada)
- 'Wrong sign?' button that saves the clip with the corrected label for retraining (active learning)
- Quick-phrase panel for emergencies (HELP, DOCTOR, PAIN) and a large-font caption mode for the listener

# 10. What Makes This Project Unique

| Feature | Why it stands out |
|---|---|
| ISL-first, not ASL | Targets an under-served language with very little tooling |
| Own open dataset with a datasheet | A reusable contribution that others can cite and extend |
| Signer-independent evaluation | Tests on people the model never saw; honest accuracy instead of inflated numbers |
| Uses face + body, not only hands | Captures non-manual markers (questions, negation, intensity) |
| Idle class + motion-based sign spotting | Works on a live stream without a 'start' button; fewer false triggers |
| ISL grammar-aware sentence builder | Converts ISL word order into fluent English instead of word-by-word output |
| Multilingual Indian speech output | Hindi and regional languages, not only English |
| Fully offline on a laptop CPU | Usable in hospitals or rural areas with no internet; protects privacy |
| Privacy by design | Only keypoints are processed/stored; no raw video leaves the device |
| Active-learning feedback loop | User corrections become new training data; model improves over time |
| Explainability view | Shows which frames / joints mattered (attention weights) and the confusable sign pairs |
| Validated with real ISL users | Feedback from a deaf school / NGO / interpreter gives real-world credibility |

If time permits, add one 'wow' feature: a **two-way mode** where the hearing person's speech is shown as large captions (speech-to-text via Whisper / IndicConformer), giving a complete conversation loop.

# 11. Evaluation Plan & Metrics

| What | Metric | Target |
|---|---|---|
| Recognition (signer-dependent) | Top-1 / Top-3 accuracy, macro F1 | at least 95% / 99% |
| Recognition (signer-independent, leave-one-signer-out) | Top-1 accuracy, macro F1, per-class recall | at least 85% |
| Confusions | Confusion matrix; list of top-5 confused pairs | Reported and explained |
| Live spotting | False triggers per minute of idle; missed signs | under 1 / min; under 10% |
| Speed | End-to-end latency, FPS on i5 CPU | under 300 ms; at least 15 FPS |
| Sentence quality | Exact-match / BLEU on 50 test gloss sequences; human rating 1-5 | at least 4 / 5 |
| Speech | Native-speaker intelligibility rating | at least 4 / 5 |
| Usability | System Usability Scale (SUS) with ISL users | at least 70 |

Ablation studies to include in the report: hands only vs. hands + pose vs. hands + pose + face; with and without normalisation; with and without augmentation; GRU vs. Transformer vs. 1D-CNN. These tables show understanding, not just a working demo.

# 12. Hardware & Software Requirements

| Hardware | Minimum | Recommended |
|---|---|---|
| Processor | Intel i5 (8th gen) / Ryzen 5 | i7 / Ryzen 7 |
| RAM | 8 GB | 16 GB |
| GPU (training only) | Not required (Google Colab / Kaggle free GPU) | Any NVIDIA GPU with 4 GB+ |
| Camera | 720p webcam | 1080p webcam, ring light |
| Audio | Laptop speaker | External speaker |

**Software:** Windows 10/11, Ubuntu 22.04 or macOS; Python 3.10/3.11; opencv-python, mediapipe, numpy, pandas, scikit-learn, torch, onnx, onnxruntime, streamlit, streamlit-webrtc, matplotlib, seaborn, IndicTrans2, Indic-TTS / Coqui TTS, pyttsx3, gTTS (optional), Git. (Pin versions in requirements.txt; MediaPipe works best on Python 3.10-3.11.)

# 13. Project Folder Structure

```
signvaani/
|-- data/
|   |-- raw_videos/            # private, not committed
|   |-- keypoints/<sign>/<signer>_<take>.npy
|   |-- splits/                # train/val/test signer lists (JSON)
|   `-- DATASHEET.md
|-- src/
|   |-- capture/record.py      # guided recording tool
|   |-- features/extract.py    # MediaPipe -> keypoints
|   |-- features/normalize.py
|   |-- data/dataset.py        # PyTorch Dataset + augmentations
|   |-- models/gru.py, transformer.py
|   |-- train.py, evaluate.py, export_onnx.py
|   |-- realtime/spotter.py    # sliding window + idle logic
|   |-- nlp/rules.json, sentence.py, translate.py
|   |-- tts/speak.py
|   `-- app/streamlit_app.py
|-- notebooks/                 # EDA, confusion matrix, ablations
|-- models/                    # .pt and .onnx checkpoints
|-- tests/
|-- requirements.txt
`-- README.md                  # setup, demo GIF, results, limitations
```

# 14. 8-Week Timeline & Team Roles

| Week | Tasks | Deliverable / Milestone |
|---|---|---|
| 1 | Finalise domain & 30-50 sign vocabulary; contact ISL user/interpreter; set up repo & MediaPipe; 5-sign proof of concept | Vocabulary list, PoC video |
| 2 | Build recording tool; record signers; download INCLUDE; keypoint extraction & normalisation; augmentation | Dataset v1 + datasheet |
| 3 | Train baseline BiGRU; signer-dependent and signer-independent splits | Baseline results |
| 4 | Transformer / 1D-CNN; ablations; confusion analysis; more data for weak signs | Model comparison table |
| 5 | Real-time spotting (sliding window, idle class, thresholds); ONNX export; latency tuning | Live recognition demo |
| 6 | Gloss-to-sentence rules + LLM polish; translation; TTS; Streamlit UI | End-to-end app |
| 7 | User testing with unseen signers & ISL user; fix failure cases; SUS survey | Test report |
| 8 | Final report, README, demo video, dataset release, presentation | Submission |

## Suggested team roles (team of 4)

| Member | Primary responsibility |
|---|---|
| A - Data lead | Recording protocol, consent, dataset, augmentation, datasheet |
| B - ML lead | Models, training, evaluation, ablations, ONNX |
| C - Real-time & NLP | Sign spotting, gloss-to-sentence, translation, TTS |
| D - App & docs | UI, integration, user testing, report, demo video |

# 15. Risks & Mitigation

| Risk | Impact | Mitigation |
|---|---|---|
| Too little data per sign | Overfitting | Augmentation, pre-train on INCLUDE, more signers rather than more takes |
| Similar signs confused | Wrong sentences | Add face/motion features; merge or replace pairs; show top-3 for confirmation |
| Hand tracking lost (fast motion, overlap) | Missing data | Mask + interpolation, hand drop-out augmentation, good lighting |
| False triggers between signs | Noisy output | Idle class, motion gate, confidence threshold, cool-down |
| Regional sign variation | Fails for some users | Document the variant used; allow user-specific fine-tuning |
| LLM hallucination | Wrong meaning spoken | Rules first; LLM limited to recognised glosses; user confirms before speaking |
| MediaPipe / Python version issues | Setup delays | Pin versions; provide a Dockerfile / conda env |

# 16. Ethics, Privacy & Accessibility

- Informed written consent from every signer; option to withdraw.
- Release only anonymised keypoints, never faces, unless explicit permission is given.

- Involve the Deaf community: the tool is an aid, not a replacement for human interpreters, especially in medical or legal settings.
- Clearly display the system's confidence and limitations in the UI.
- Design UI with high contrast, large text and simple icons.

## 17. Expected Outcomes & Deliverables

- Working offline desktop app: ISL sign -> sentence -> speech in 2+ languages
- Public ISL keypoint dataset (30-50 signs, 6+ signers) with datasheet and licence
- Trained models (.pt and .onnx) and reproducible training code on GitHub
- Report with signer-independent results, ablations, confusion analysis and limitations
- 3-minute demo video and user-feedback summary

## 18. Future Scope

- Continuous sign language translation with CTC / sequence-to-sequence models trained on iSign
- Speech-to-ISL using a 3D avatar for two-way conversation
- Android app with TFLite / MediaPipe on-device
- Fingerspelling recognition for names and places
- Extending to more domains (railway, banking, education) and regional ISL variants
- Federated / on-device personalisation for each user

## 19. Conclusion

SignVaani demonstrates that a practical, privacy-preserving ISL-to-speech assistant can be built with commodity hardware by combining landmark-based perception, compact sequence models and Indian-language speech technology. By focusing on a realistic domain vocabulary, contributing an open dataset and reporting signer-independent performance, the project offers both a usable prototype and a genuine contribution to low-resource sign language research.

## 20. References


- Sridhar, A., Ganesan, R. G., Kumar, P., Khapra, M. (2020). INCLUDE: A Large Scale Dataset for Indian Sign Language Recognition. ACM Multimedia.
- Selvaraj, P., NC, G., Kumar, P., Khapra, M. (2022). OpenHands: Making Sign Language Recognition Accessible with Pose-based Pretrained Models across Languages. ACL.
- Elakkiya, R., Natarajan, B. (2021). ISL-CSLTR: Indian Sign Language Dataset for Continuous Sign Language Translation and Recognition. Mendeley Data.
- Joshi, A., et al. (2024). iSign: A Benchmark for Indian Sign Language Processing. Findings of ACL.
- Lugaresi, C., et al. (2019). MediaPipe: A Framework for Building Perception Pipelines. arXiv:1906.08172.
- Gala, J., et al. (2023). IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages. TMLR.
- Kumar, G. K., et al. (2023). Towards Building Text-to-Speech Systems for the Next Billion Users (Indic-TTS). ICASSP.

- Bohacek, M., Hruz, M. (2022). Sign Pose-based Transformer for Word-level Sign Language Recognition (SPOTER). WACV Workshops.
- Li, D., et al. (2020). Word-level Deep Sign Language Recognition from Video: A New Large-scale Dataset (WLASL). WACV.
- Gebru, T., et al. (2021). Datasheets for Datasets. Communications of the ACM.
- Indian Sign Language Research and Training Centre (ISLRTC) - ISL Dictionary, Government of India.

# Appendix A: Suggested Vocabulary - Hospital Domain (40 signs + IDLE)

| Category | Signs |
|---|---|
| People | ME / I, YOU, DOCTOR, NURSE, FAMILY |
| Symptoms | PAIN, FEVER, COUGH, VOMIT, DIZZY, BLEEDING, BREATHING-PROBLEM, ALLERGY |
| Body parts | HEAD, STOMACH, CHEST, TOOTH, EAR, EYE, LEG, HAND |
| Needs / actions | HELP, WATER, MEDICINE, TOILET, FOOD, SLEEP, CALL, WAIT, INJECTION |
| Questions / grammar | WHERE, WHAT, WHEN, HOW-MUCH, YES, NO, NOT |
| Time / intensity | TODAY, YESTERDAY, VERY / MORE |
| Special | IDLE / NO-SIGN (mandatory) |

## Example sentences the system should handle

| ISL gloss | English output | Hindi output |
|---|---|---|
| ME PAIN STOMACH | I have stomach pain. | Mere pet mein dard hai. |
| ME FEVER YESTERDAY | I have had a fever since yesterday. | Mujhe kal se bukhar hai. |
| TOILET WHERE | Where is the toilet? | Shauchalay kahan hai? |
| ME NEED WATER | I need water. | Mujhe paani chahiye. |
| DOCTOR CALL | Please call the doctor. | Kripya doctor ko bulaiye. |
| ME ALLERGY MEDICINE | I am allergic to a medicine. | Mujhe dawa se allergy hai. |

*Hindi shown in Latin transliteration here; the app displays and speaks Devanagari. Verify every sign and translation with an ISL user and native speaker. Alternative domains: railway station (TICKET, PLATFORM, TRAIN, LATE...), bank (ACCOUNT, MONEY, DEPOSIT...), college (CLASS, EXAM, LIBRARY...).*

# Appendix B: Likely Viva / Review Questions

- Why keypoints instead of raw video? (speed, data efficiency, background robustness, privacy)
- How do you normalise for different signer positions and body sizes?
- What is signer-independent evaluation and why does accuracy drop?
- How does the system know when one sign ends and the next begins?
- Why is an IDLE class needed?
- Why GRU vs. Transformer for this data size? What did your ablation show?
- How is ISL grammar different from English, and how do you handle it?
- Which signs are confused most, and why?
- How would you scale to continuous sentences?
- What are the ethical considerations of releasing a sign language dataset?